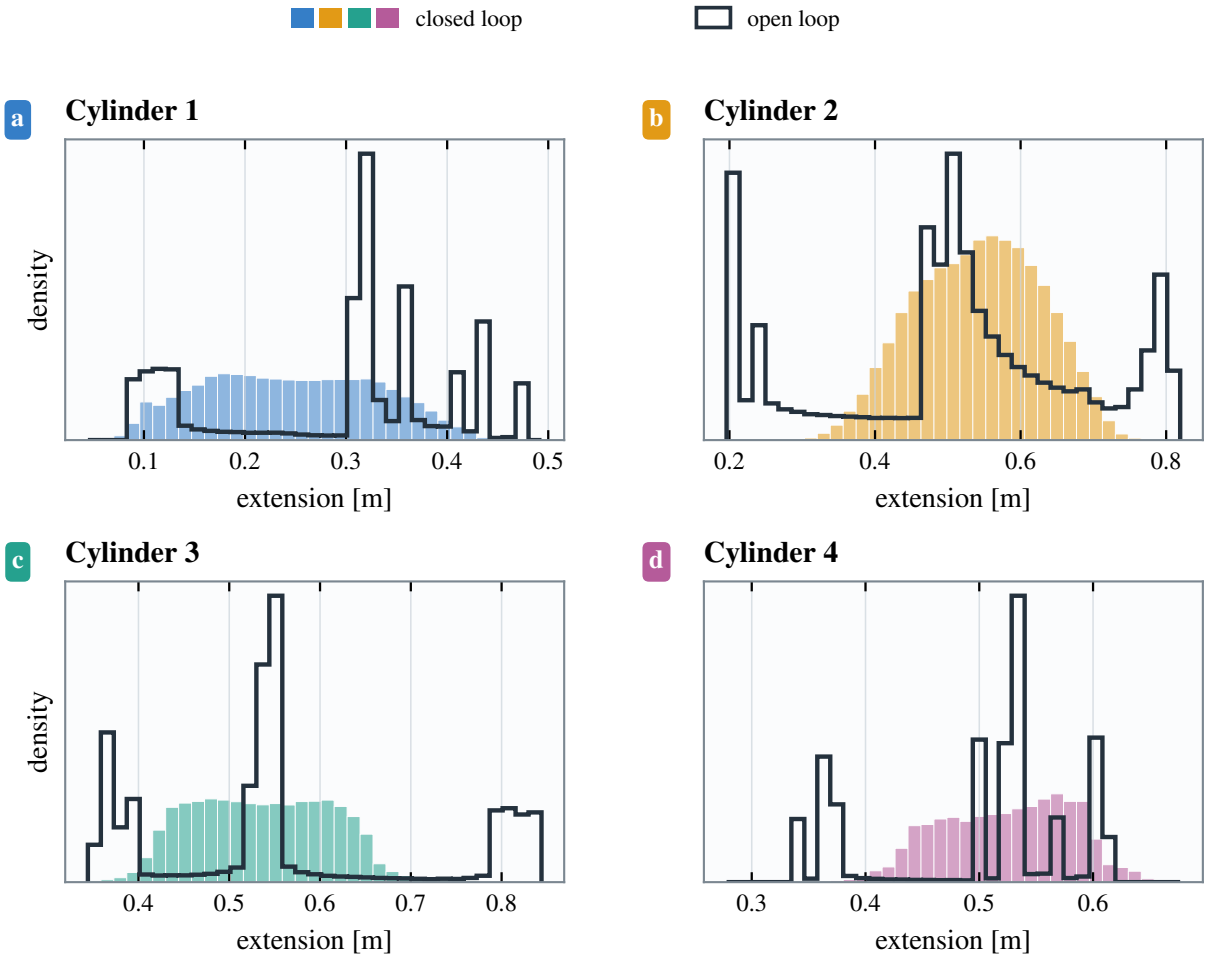


Figure 4.4: Extension coverage of the two campaigns, per cylinder and pooled over runs (closed-loop filled, open-loop outline). The closed-loop data fills the reachable range smoothly the open-loop data concentrates into the plateaus where the valve is identifiable.

define its reproducibility boundary. Exact alignment, increment consistency and the identifiability failure were checked directly on the export.

The separate thousand-run randomized curriculum was collected for an earlier adaptation plan. No reported result is trained on or evaluated against it; the adaptation chain uses the IsaacLab collection of Section 4.7. The curriculum is retained as provenance and documented in Appendix B.

4.6 Closed-Loop Calibration Data for the Command ROM

The ROM drives a PhysX articulation whose compliance and contact response are not present in Simscape. Further Simscape training cannot remove the resulting deployment offset.